Hints on Evaluating Quantile using the MELE Method (HW04 P1)

Recall the setup for our MELE problem. We have an N measurements of a random variable x, which can take on integer values x_i. We sort the measurements into bins of width 1. Next, we take the contents of each bin n_i and divide it by N to obtain an empirical probability distribution p_i, with the property that they sum to unity:

$$p_i = \frac{n_i}{N}, \qquad \sum_{i=1}^{N} p_i = 1$$

In applying the MELE method, we incorporate the quantity of interest into a "constraint" applied to the x_i and their corresponding p_i. For example, if we're interested in the mean we write the constraint as

$$\sum_{i=1}^{N} p_i(x_i - \mu) = 0$$

and define an "estimating function" for our code as follows:

$$g(x, \mu) = x - \mu$$

Similarly, for a lab exercise we studied the variance σ^2 of a distribution (for which the ground-truth mean μ_{gt} was known) by writing the constraint as

$$\sum_{i=1}^{N} p_i \left[(x_i - \mu_{gt})^2 - \sigma^2 \right] = 0$$

and defining an estimating function

$$g(x, \sigma^2) = (x - \mu_{gt})^2 - \sigma^2$$

Now a quantile α, where $0 < \alpha < 1$, is a particular fraction of the data. The statement "x_α is the α quantile of the data" means that α is the fraction of the data for which $x < x_\alpha$. How do we express this in the form of a constraint? What we want is to sum the p_i only for those x_i which are less than x_α. Something like:

$$\sum_{i=1}^{N} p_i \left[S(x_i) - \alpha \right] = 0$$

where $S(x_i)$ is a step function equal to

$$\begin{aligned} S(x_i) &= 1 \text{ if } x_i < x_\alpha \\ &= 0 \text{ if } x_i \geq x_\alpha \end{aligned}$$

There are both clever and not-so-clever ways of implementing $S(x_i)$ in Python.